

1000 BCE ▶ 1 CE

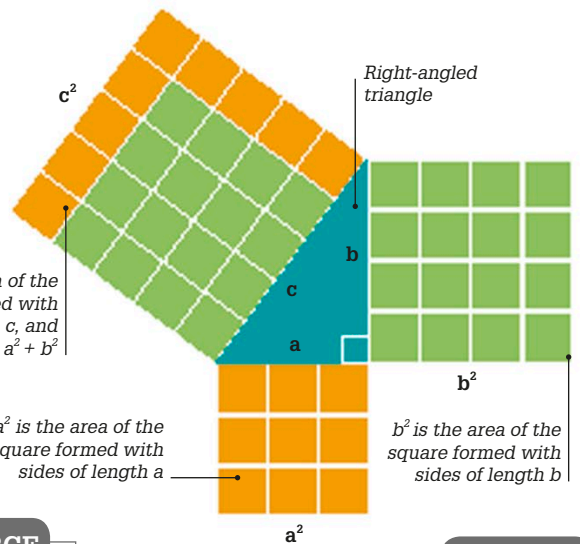


Around 450 BCE, Empedocles of Acragas (a Greek colony in Sicily) had the idea that all matter is made up of four basic elements: earth, air, fire, and water.



600 BCE Oldest world map

The oldest-known attempt to create a world map was made on a clay tablet in Babylon (now modern Iraq). The tablet portrays the world as a flattened disc, surrounded by an ocean. Babylon is shown as a rectangle in the center, with eight other cities indicated by circles.



530 BCE Pythagorean theorem

Greek mathematician Pythagoras was interested in the mystical powers of numbers. A version of the theorem named after him was known to the Egyptians and Babylonians, but Pythagoras was the one who worked it out. It states that the sum of the squares of the two shorter sides of a right-angled triangle is equal to the square of the longer side.



1000

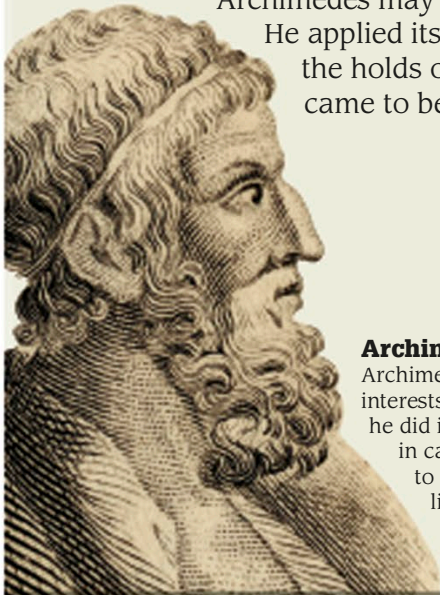
800

600

700 BCE ARCHIMEDES SCREW PUMP

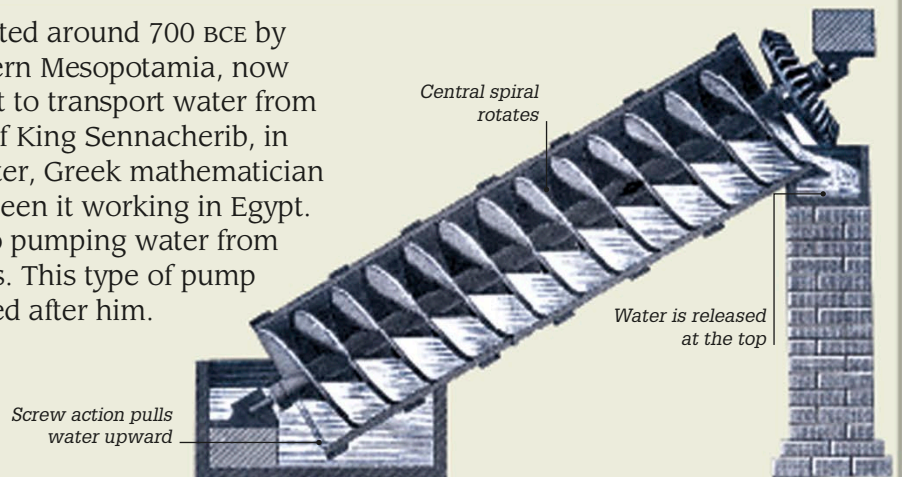
The screw pump was probably invented around 700 BCE by the Assyrians (people living in northern Mesopotamia, now in modern Iraq). These people used it to transport water from one level to another in the gardens of King Sennacherib, in their capital of Nineveh. Centuries later, Greek mathematician Archimedes may have seen it working in Egypt.

He applied its use to pumping water from the holds of ships. This type of pump came to be named after him.



Archimedes of Syracuse

Archimedes (c 287–212 BCE) had a vast range of interests. As well as developing the screw pump, he did important work on geometry, especially in calculating the area of a circle. He is said to have invented a heat ray by focusing light on an array of mirrors.



How the screw works

Water enters an Archimedes screw from the bottom. When the central spiral of the screw is rotated, water is pulled through it and transferred to a higher level, from where it exits the pump.